

Homework 2: (Return by Friday 3.10.25 in PDF -format)

A power MOSFET transistor can be thermally modelled with the thermal circuit of Fig. 1. The MOSFET is pulsed with a burst of 5 ms wide (FWHM), 75 W (heat power) pulses. How large can the pulsing frequency be so that the MOSFET does not break down (max T junction 125 °C) in room temperature (20 °C) conditions after a pulse burst of $\text{floor}[3+(n/3)]$ pulses? Floor refers to a mathematical function (more on Wikipedia, for example) and n is the last digit of your student number. The solution should include the thermal circuit and the simulation result that fulfils and demonstrates the above requirements. **Determine also without simulations what is the maximum possible power dissipation (DC power, NOT pulsed) with this transistor?** Also list three things that have an effect on the power dissipation within a power MOSFET-transistor.

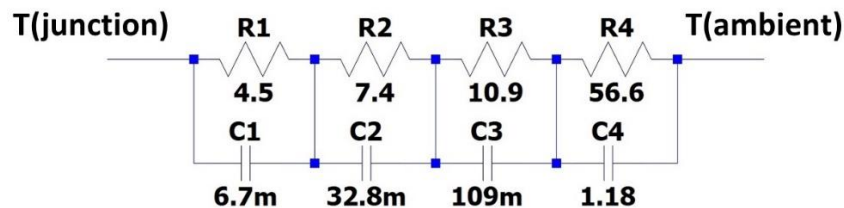


Fig. 1 A thermal model of the power MOSFET.

The simulations can be done with the simulator found in <https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>. You can find quick guides to LTspice online. Questions to jaakko.huikari@oulu.fi.

Important practical notes on your return documents on this course:

- Name your return file in the following manner: ESD25 HW#N first name surname. As an example, in the case of homework 2 => ESD25 HW#2 Jaakko Huikari (first- /surname matching your name in the course's Moodle page (Enrolled users)).
- Return your documents in **PDF -format**. Do not return documents in Word- or other formats. Convert your word-document, for example, into PDF -format before you return it.
- Place your name and student ID number at the top edge of your return document. Use the name matching your name in the course's Moodle page (Enrolled users).
- Make sure your answers and simulation results are clearly visible in your return documents. If necessary, use cursors, zoom and screen captures to present sufficient graphic information (waveforms etc.). Make sure that figure axes are also visible in your return documents.
- Present the reasoning behind your answers/ results including theoretical basis and equations etc. (found in the lecture material). Presenting mere outcome is inadequate answer.